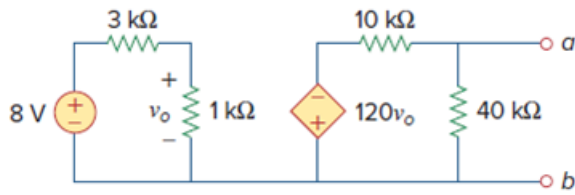


Problem

For the circuit in Fig. 4.137, what resistor connected across terminals a - b will absorb maximum power from the circuit? What is that power?

Figure. 4.137:



Step-by-step solution

Step 1 of 7

We need to find V_{TH} and I_{SC} at terminals a and b . To find R_{TH} ,

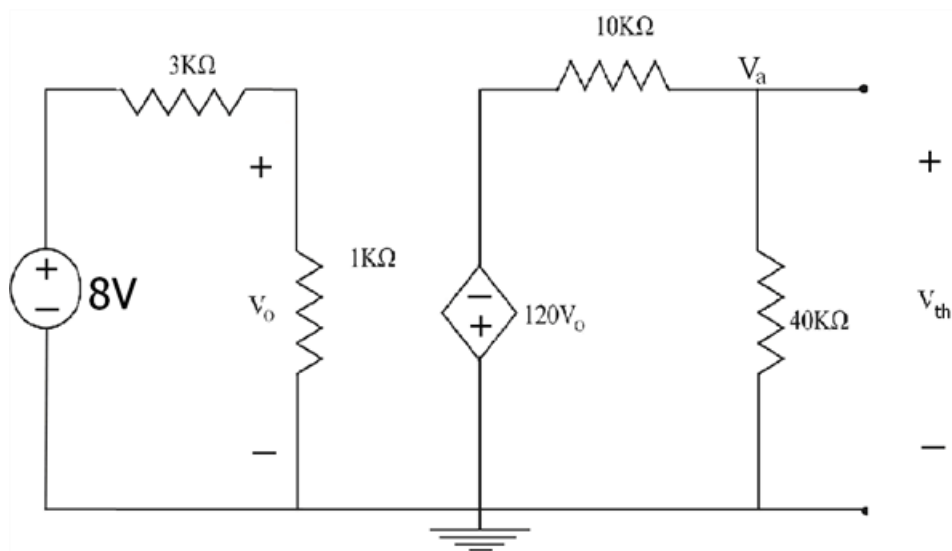


Fig-(a)

Step 2 of 7

Assume that all the resistances are in $k\Omega$, all currents are in mA, and all voltages are in volts.

At node 'a' by applying KCL we can get

$$\frac{V_a + 120V_o}{10} + \frac{V_a}{40} = 0$$

$$V_a \left[\frac{1}{10} + \frac{1}{40} \right] + 12V_o = 0$$

$$V_a \left[\frac{1}{8} \right] + 12V_o = 0 \dots \dots \dots (1)$$

Step 3 of 7

Where from the left part of the circuit

$$V_0 = 8 \left(\frac{1}{3+1} \right)$$

$$V_0 = 2\text{V}$$

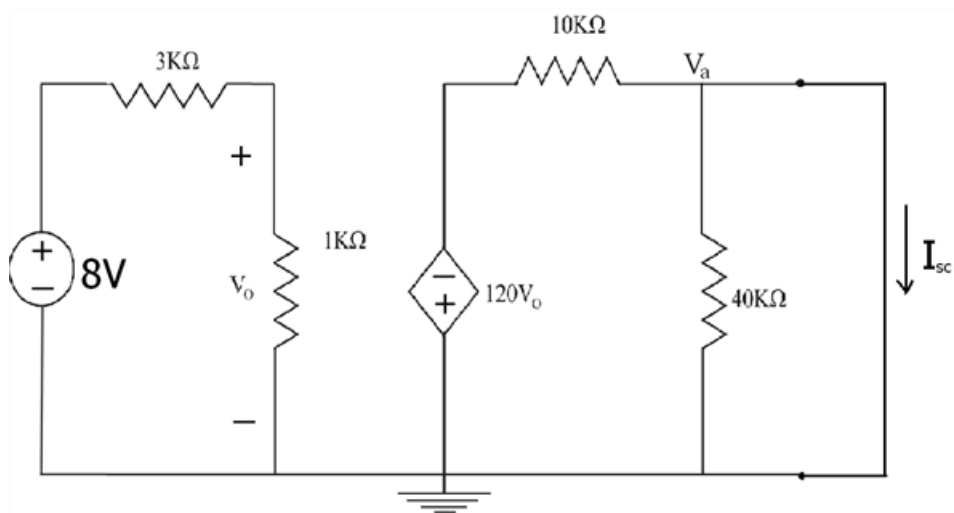
From equation (1)

$$V_a = -192\text{V}$$

Hence the thevenin voltage is

$$V_{TH} = -192\text{V}$$

Step 4 of 7



Step 5 of 7

Since the terminals ab are short circuited, no current flows through $40k\Omega$

$$I_{sc} = -\frac{120V_0}{10k}$$

Where from the left part of the circuit

$$V_0 = 8 \left(\frac{1}{3+1} \right)$$

$$V_0 = 2\text{V}$$

Hence,

$$I_{sc} = -\frac{120(2)}{10k}$$

$$I_{sc} = -24\text{mA}$$

Step 6 of 7

Thevenin's equivalent resistance is,

$$R_{TH} = \frac{V_{th}}{I_{sc}}$$

$$R_{TH} = \frac{-192}{-24 \times 10^{-3}}$$

$$R_{TH} = 8\text{k}\Omega$$

Hence for maximum power transfer the resistor to be connected is equal to the thevenin resistance, $R_{ab} = 8\text{k}\Omega$

Step 7 of 7

Hence the maximum power transferred is

$$P_{\max} = \frac{\left[V_{TH} \left(\frac{8\text{k}}{8\text{k}+8\text{k}} \right) \right]^2}{R_{TH}}$$

$$P_{\max} = \frac{V_{TH}^2}{(4 R_{TH})}$$

$$P_{\max} = \frac{(-192)^2}{(4 \times 8 \times 10^3)}$$

$$P_{\max} = 1.152 \text{ watts}$$